

Assessment Case Study : Mutual Fund Portfolio Performance & Risk Analysis

Business Scenario

A leading Asset Management Company (AMC) manages over **5 million investor transactions** across multiple mutual fund schemes. The Fund Management team wants to analyze portfolio performance, identify high-risk investors, evaluate fund returns, and generate automated investment reports.

Input Files

investors.csv

funds.csv

transactions.csv

nav_history.csv

Requirements

Python

- Read multiple CSV files
 - Handle missing or corrupted files using Exception Handling
 - Create reusable functions
 - Implement OOP using a **FundPortfolio** class
 - Use logging for report generation
 - Generate reports automatically
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NumPy

Calculate

- ✓ Mean Investment Amount
 - ✓ Median Investor Income
 - ✓ Standard Deviation of NAV
 - ✓ Percentile Fund Returns (90th & 95th)
 - ✓ Correlation between Investor Income and Investment Amount
 - ✓ Average Daily NAV
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Pandas

Merge

- ✓ investors
 - ✓ transactions
 - ✓ funds
 - ✓ nav_history
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Identify

Top 20 Investors based on Portfolio Value

Investors having

- ✓ Investment > ₹10 Lakhs
 - ✓ High Risk Profile
 - ✓ More than 10 Transactions
 - ✓ Annual Income > ₹15 Lakhs
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Fund Analysis

Find

- ✓ Best Performing Fund
 - ✓ Worst Performing Fund
 - ✓ Highest Expense Ratio
 - ✓ Highest AUM (Assets Under Management)
 - ✓ Most Popular Fund
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Missing Data

Replace

- ✓ Annual Income → Median
 - ✓ Expense Ratio → Mean
 - ✓ NAV → Previous Day NAV
 - ✓ Risk Profile → "Moderate"
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Outlier Detection

Remove

- ✓ Investment Amount > 99th Percentile
- ✓ NAV Changes > 3 Standard Deviations

Finance Metrics

Calculate

- ✓ Total Portfolio Value
- ✓ Portfolio Return %
- ✓ CAGR (Compound Annual Growth Rate)
- ✓ Absolute Return
- ✓ Annualized Return
- ✓ Portfolio Diversification Score
- ✓ Average Holding Period
- ✓ Expense Ratio Impact
- ✓ Sharpe Ratio (Simplified)
- ✓ Category-wise Investment %
- ✓ Fund Allocation %
- ✓ Investor-wise Profit/Loss

Data Visualization

Generate Charts

- ✓ Portfolio Allocation (Pie Chart)
- ✓ Fund-wise Investment (Bar Chart)
- ✓ Monthly Investment Trend (Line Chart)
- ✓ Category-wise Returns (Bar Chart)
- ✓ NAV Movement (Line Chart)
- ✓ Top 10 Investors (Horizontal Bar Chart)

Mini Project

Automated Mutual Fund Performance Dashboard

The script should:

- ✓ Read all CSV files
- ✓ Clean missing values
- ✓ Remove duplicate transactions
- ✓ Calculate portfolio metrics
- ✓ Rank funds

✓ Identify high-value investors

✓ Create charts

✓ Export reports

✓ Log execution status
